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Physiologically Based Pharmacokinetic Model to Predict Drug–Drug Interactions With the Antibody–Drug Conjugate Trastuzumab Deruxtecan

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ABSTRACT

The topoisomerase I inhibitor payload (DXd) released from the human epidermal growth factor receptor 2-targeted antibody–drug conjugate (ADC) trastuzumab deruxtecan (T-DXd) is eliminated by hepatic uptake via OATP1B1/3, metabolism by CYP3A, biliary excretion via P-glycoprotein and breast cancer resistance protein, and urinary excretion. In a clinical drug–drug interaction (DDI) study between T-DXd and ritonavir (a strong cytochrome P450 [CYP] 3A/organic anion transporting polypeptide 1B inhibitor) or itraconazole (a strong CYP3A inhibitor), a 1.2-fold increase in DXd exposure was observed. In this study, a physiologically based pharmacokinetic (PBPK) model of T-DXd was constructed to quantitatively describe the pharmacokinetics (PK) of DXd and the DDI of DXd released from T-DXd. The minimal ADC PBPK model was developed to describe both intact ADC and payload PK using in vitro data of DXd and clinical PK data after T-DXd administration. The developed model reasonably described the PK of T-DXd and DXd after intravenous administration of T-DXd. The main elimination pathway of DXd was estimated to be biliary excretion, followed by urinary excretion, with metabolism contributing to a lesser extent. DDIs with ritonavir and itraconazole simulated by the PBPK model reproduced the clinical DDI study results, indicating that the impact of concomitant medications was not clinically meaningful. These findings suggest that this model may potentially be used for DDI evaluation of future DXd-containing ADCs, aiding in the prediction of DDIs under different scenarios and in specific populations.

1 | Introduction

Trastuzumab deruxtecan (T-DXd) is a human epidermal growth factor receptor 2 (HER2)-targeted antibody–drug conjugate (ADC) composed of a humanized monoclonal antibody specifically targeting HER2 (trastuzumab), a cleavable tetrapeptide-based linker, and DXd, a potent topoisomerase I inhibitor payload molecule [1, 2]. T-DXd is under development for the treatment of solid tumors, including, but not limited to, breast cancer, gastric cancer, and non-small cell lung cancer. The approved dose of T-DXd is 5.4 mg/kg every 3 weeks for breast cancer and 6.4 mg/kg every 3 weeks for gastric cancer.

Patients with cancer are often elderly; they may be receiving concomitant medications for other diseases and treatments for infections, which raises concerns about potential drug–drug interactions (DDIs). ADCs consist of an antibody component linked to a payload, necessitating an evaluation of DDIs involving both the antibody and the payload [3]. However, reports on DDIs related to the antibody component are generally limited. In contrast, the payload is a small molecule that can function as a substrate or inhibitor/inducer of metabolic enzymes and transporters, requiring a similar assessment of DDI risks similar to those of conventional small molecules. Regarding the perpetrator DDI risk of ADC payloads in general, while in vitro

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Study Highlights

- What Is the Current Knowledge on the Topic?
 - Antibody–drug conjugates (ADCs) can present drug–drug interaction (DDI) risks through small-molecule payloads. DXd, a potent payload of trastuzumab deruxtecan (T-DXd), is mainly eliminated by biliary excretion, with contributions from renal clearance and cytochrome P450 3A-mediated metabolism.
- What Question Did This Study Address?
 - This study demonstrates how physiologically based pharmacokinetic (PBPK) modeling can characterize the pharmacokinetics of DXd released from T-DXd and informs DDI risk assessment for subsequent DXd-containing ADCs.
- What Does This Study Add to Our Knowledge?
 - The model successfully reproduced clinical pharmacokinetic data of T-DXd and DXd, demonstrating minimal increases in DXd exposure with ritonavir and itraconazole. Sensitivity analyses confirmed the suitability of the elimination parameters used in the DXd model.
- How Might This Change Drug Discovery, Development, and/or Therapeutics?
 - The T-DXd PBPK model can be applied to predict DDIs and pharmacokinetics in future DXd-containing ADCs.

evaluations have shown inhibitory or inductive effects on metabolic enzymes and transporters, the actual risk of DDIs may be limited because of the very low plasma concentrations of the payload. Regarding victim DDIs, because the payload is generally cytotoxic and its plasma concentration often affects safety, concerns may arise if the payload serves as a substrate for metabolic enzymes or transporters, as co-administration with inhibitors of these elimination pathways could lead to increased plasma concentrations.

In vitro studies have identified DXd, the payload of T-DXd, as a substrate of cytochrome P450 (CYP) 3A, organic anion transporting polypeptide (OATP) 1B1/1B3, P-glycoprotein (P-gp), and breast cancer resistance protein (BCRP) [4]. Furthermore, excretion studies following intravenous administration of [^{14}C]T-DXd in monkeys demonstrated that the radiolabeled compound was primarily excreted in feces, with some payload excreted in urine. The main component detected in both urine and feces was unchanged DXd. In bile-duct cannulated rats, administration of [^{14}C]DXd showed that most of the radioactivity was excreted in bile, with unchanged DXd being the main component detected [4]. These findings suggest that DXd is predominantly excreted unchanged via bile, with partial metabolism by CYP3A. Therefore, concomitant use of CYP3A inhibitors or transporter inhibitors involved in biliary excretion may lead to increased plasma concentrations of DXd.

In clinical settings, although the plasma concentration of DXd after T-DXd administration is much lower than that of intact

T-DXd, it has been shown to be associated with hematologic toxicity [5]. Thus, it is important to evaluate the impact of concomitant medications on the pharmacokinetics (PK) of DXd. A clinical DDI study was conducted with T-DXd, assessing the effects of ritonavir, which has inhibitory effects on both CYP3A and OATP1B, and itraconazole, a strong CYP3A inhibitor [6]. When T-DXd was administered intravenously at 5.4 mg/kg in combination with either ritonavir or itraconazole, the serum concentration of DXd increased by approximately 1.2-fold in both cases. Moreover, combination treatment did not result in clinically significant safety changes compared with T-DXd monotherapy. It was concluded that the increase in DXd exposure resulting from the co-administration of ritonavir or itraconazole was not clinically significant.

Currently, multiple ADCs using the same DXd payload are under development. The use of physiologically based pharmacokinetic (PBPK) modeling for the assessment of DDI risk of ADCs with the same payload has been proposed [3]. PBPK modeling has been used for DDI risk assessments of a series of ADCs containing monomethyl auristatin E (MMAE) as a payload. MMAE, the payload of brentuximab vedotin, polatuzumab vedotin, and enfortumab vedotin, is a substrate of CYP3A and P-gp that acts as a CYP3A inhibitor [7]. Clinical DDI studies have been conducted for brentuximab vedotin to evaluate the DDI risks both as a perpetrator and as a victim for its payload, MMAE. Subsequently, PBPK modeling was employed for DDI evaluations of polatuzumab vedotin and enfortumab vedotin [8–10]. The PBPK model construction incorporated in vitro studies of MMAE and ADCs containing MMAE, in vivo absorption, dissolution, metabolism, and excretion (ADME) study results in animals, and clinical PK profile data. Furthermore, the PBPK model was validated using clinical DDI data of brentuximab vedotin. These PBPK models were used to predict DDIs with concomitant medications following administration of ADCs containing MMAE, with the results reflected in the product labeling. Thus, when multiple ADCs with the same payload are under development, existing DDI study data from earlier ADCs can be very useful in assessing DDI risks for subsequent ADCs using PBPK models.

In this study, we incorporated the existing PK properties of DXd into a PBPK model to gain a deeper understanding of the PK characteristics of DXd following T-DXd administration and to assist in the DDI risk assessment for subsequent DXd-ADCs.

2 | Methods

2.1 | Clinical Data

Clinical data from studies DS8201-A-J101 and DS8201-A-A104 were used in this modeling exercise [6, 11]. In Clinical Study DS8201-A-J101, T-DXd was administered at 5.4 and 6.4 mg/kg every 21 days in the dose-expansion cohort [11]. Data from the dose-expansion cohorts (Parts 2a and 2c) were used for the PBPK modeling. In Clinical Study DS8201-A-A104, T-DXd was administered at 5.4 mg/kg every 21 days [6]. The DDI assessment was conducted during Cycles 2 and 3. Inhibitors were administered from Day 17 of Cycle 2 until the end of

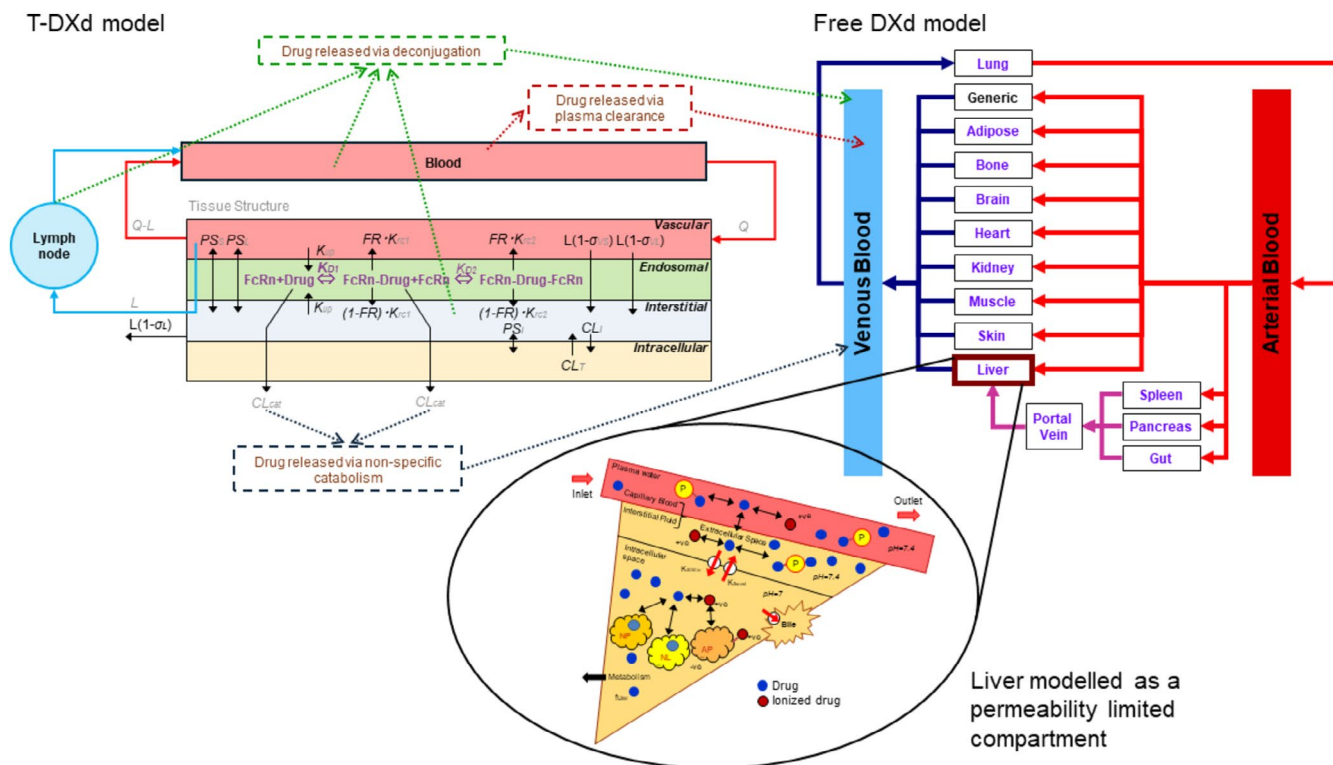


FIGURE 1 | Minimal PBPK model for ADCs including 2:1 binding to FcRn at pH 6.0 in the endosomal compartment linked to a full PBPK model for released DXd with a nested permeability limited liver model. P represents plasma protein; +ve and -ve represent the drug in an ionized form, that is, with and without a valence electron. Figure adapted from Jamei et al. [13] and Huang et al. [14]. ADC, antibody–drug conjugate; AP, acidic phospholipids; CL_T , clearance of drug from interstitial to intracellular compartment; CL_{cat} , catabolic clearance; CL_T , clearance of drug from intracellular to interstitial compartment; FcRn, neonatal Fc receptor; FR, fraction of FcRn-bound monoclonal antibody recycled; $f_{u,iv}$, fraction unbound in the intracellular water; K_{D1} , equilibrium dissociation constant for 1:1 drug–FcRn complex; K_{D2} , equilibrium dissociation constant for 2:1 drug–FcRn complex; K_{rc1} , endosomal recycling rate for 1:1 drug–FcRn complex; K_{rc2} , endosomal recycling rate for 2:1 drug–FcRn complex; K_{up} , endosomal uptake rate via fluid phase endocytosis; L, lymph flow rate; NL, neutral lipids; NP, neutral phospholipids; PBPK, physiologically based pharmacokinetic; PS_s , permeability surface area product for small pores; PS_l , permeability surface area product for large pores; PS_t , permeability surface area product for tissue permeability of drug; Q, blood flow rate; T-DXd, trastuzumab deruxtecan; σ_l , lymphatic reflection coefficient; $\sigma_{v,s}$, vascular reflection coefficient for vascular permeability through small pores; $\sigma_{v,l}$, vascular reflection coefficient for vascular permeability through large pores.

Cycle 3. Ritonavir was administered at a dose of 200 mg twice daily (BID), and itraconazole at 200 mg BID on Day 17 of Cycle 2 and then 200 mg once daily on subsequent days. The effect of the inhibitors on payload kinetics was assessed by comparing the area under the curve (AUC)_{17day} and maximum plasma concentration (C_{max}) in Cycle 2 (no inhibitor) and Cycle 3 (with inhibitor).

2.2 | PBPK Model

The Simcyp Population-Based Simulator Version 21 (Certara UK Limited, Sheffield, UK) was used for all simulations. Serum and plasma concentrations were assumed to be equivalent.

2.3 | T-DXd Model

The minimal ADC model within the Simcyp Simulator was used to describe the PK of T-DXd. In the model, extravasation of the protein from the vascular space into the tissue, endosomal uptake, binding to neonatal Fc receptor (FcRn), and subsequent recycling to the vascular space or transcytosis to the interstitial

space are accounted for [10, 12]. The structure of the minimal PBPK model for ADCs is shown in Figure 1. Distribution of the ADC into the intracellular compartment was not considered in the model for T-DXd.

The equilibrium dissociation constant for 1:1 drug–FcRn complex (K_{D1}) for T-DXd was calibrated using the measured data for trastuzumab ($K_D = 1.64 \mu M$) (data on file) and published data for trastuzumab ($K_D = 0.773 \mu M$ [15]), which came from the same publication as the endogenous immunoglobulin (Ig) G K_D used in the Simcyp Simulator [12]. A 1:2 stoichiometry for the binding of ADC and FcRn was modeled as this is more physiologically realistic than the use of a 1:1 binding stoichiometry [16].

The endosomal uptake rate via fluid phase endocytosis (K_{up}) and fraction of FcRn-bound monoclonal antibody recycled (FR) values were estimated using the parameter estimation module within the Simcyp Simulator by fitting to the observed plasma concentrations following 5.4 mg/kg (Clinical Study DS8201-A-A104) [6, 17]. The catabolic clearance (0.0175 L/h) and additional clearance (0.0017 L/h) values were retained at the values used to describe endogenous IgG kinetics in the Simcyp Simulator.

Within the ADC model, the payload can be liberated from the ADC by sequential deconjugation or, if an ADC molecule is cleared (i.e., via catabolic clearance or degradation of the ADC), all of the payload still conjugated to the ADC is liberated. The default sequential deconjugation rate constant in Simcyp V21 ($K_{\text{dec}} = 0.004167 \text{ h}^{-1}$) was used for T-DXd. Deconjugation had a minimal contribution to the release of DXd in the simulations. Complete degradation of the ADC in the endosome was the main driver for DXd appearance in plasma in the PBPK model. Following ADC degradation, all payload molecules can be fully released into plasma or partially released, with only a fraction of the payload molecules entering the circulation. The fraction of payload released from the site of catabolism (F_{rel}) in the PBPK model was set to 0.5, as this resulted in concentrations of DXd that were in line with the observed data. The final input parameters for the T-DXd ADC PBPK model are summarized in Table 1.

2.4 | DXd Model

The PBPK model for DXd was developed using a bottom-up approach and available in vitro and in vivo non-clinical data. The steady-state volume of distribution of DXd was predicted to be 0.2 L/kg using Method 3 [22] within the Simcyp Simulator. This is similar to the allometrically scaled value from monkey data (0.15 L/kg) [4], after accounting for interspecies differences in unbound fraction. The clearance of DXd is dependent on metabolism, hepatic uptake, hepatic efflux, and renal clearance processes. To accommodate all of these processes, a permeability-limited model of the liver (PerL) was nested in the full PBPK model for DXd [13].

In the PerL model, the rate of metabolism of DXd was calculated from the unbound intrinsic metabolic clearance (CL_{int}) scaled from human liver microsomes (HLM). The HLM CL_{int} was divided between CYP3A4 clearance and other metabolic pathways based on an in vitro HLM ketoconazole experiment.

The passive diffusion (CL_{PD}) and the overall active hepatic uptake of DXd were calculated based on the results of in vitro human hepatocyte experiments. Data from HEK293 cells expressing OATP transporters were used to estimate the ratio of the OATP1B1/OATP1B3 contribution to DXd hepatic uptake, assuming the same expression levels of OATP1B1 and OATP1B3 in transfected HEK cells. The rate of biliary efflux was calculated from the scaled unbound hepatic P-gp and BCRP intrinsic clearances from human colon adenocarcinoma (Caco-2) cells and the relative expression factor for P-gp and BCRP between Caco-2 cells and hepatocytes. A detailed description of the in vitro modeling of P-gp, BCRP, OATP1B1, and OATP1B3 and derivation of their transport kinetic parameters is available in Methods S1 and Table S1.

It was assumed that any DXd effluxed into bile in the PerL model entered the intestine via gall bladder emptying, where it was excreted unchanged into the feces rather than re-absorbed. This assumption is consistent with findings in non-clinical ADME studies, where the compound is largely excreted unchanged in feces [4]. This is assumed to be due to the action of intestinal efflux transporters (BCRP and P-gp).

The renal clearance was predicted to be 2.82 L/h based on allometric scaling using monkey renal clearance data [4], after accounting for interspecies differences in unbound fraction. The final input parameters for the DXd PBPK model are summarized in Table 1.

2.5 | Model Validation

The PK of T-DXd and DXd and the DDIs with ritonavir and itraconazole were simulated and compared with observed clinical data from Clinical Study DS8201-A-A104 [6]. The default PBPK models for ritonavir and itraconazole (and OH-itraconazole) within Simcyp V21 were used in DDI simulations. In this study, the reproducibility of the inhibitor plasma exposures (ritonavir and itraconazole) could not be confirmed because the inhibitor plasma concentrations were not measured. However, the predictive accuracy of these models has been confirmed using clinical data in Simcyp. The default ritonavir model does not contain inhibition constant (K_i) values for OATP1B1 and OATP1B3; hence, values of $0.042857 \mu\text{M}$ and $0.051429 \mu\text{M}$ were added into the ritonavir PBPK model (further details are given in the Methods S2, Tables S2 and S3). PK prediction performance was assessed by comparing predicted and observed PK parameters. A predicted-to-observed ratio between 0.5 and twofold was considered acceptable based on published criteria [23].

The DDI simulations were conducted with the Sim-Japanese and Sim-Cancer populations. Simulations were conducted using 10 trials per study subject and were aligned with the study design. Where possible, the demographics of the simulated population were matched to those of the subjects in the clinical trials. For the ritonavir study, the simulations were performed with 10 trials of 12 Japanese subjects aged 48–70 years or cancer patients aged 48–80 years (67% were female), based on Clinical Study DS8201-A-A104 [6]. For the itraconazole cohort, there were 14 Asian subjects aged 31–69 years (28.6% were female).

2.6 | DXd Pathway Mass Balance

To verify whether the DXd model—constructed using a bottom-up approach based on in vitro and preclinical in vivo data—accurately describes the mass balance of DXd, pathway mass balance was assessed following a single intravenous (IV) dose of DXd (0.1 mg administered over 0.5 h). The simulations were performed with 10 trials of 14 cancer patients aged 31–69 years (28.6% were female), based on the design of Clinical Study DS8201-A-A104 [6].

2.7 | Accounting for Risk of Uncertainty in DXd DDI Predictions

Owing to the lack of clinical data following direct administration of DXd to verify the DXd model parameters, sensitivity analyses were performed to assess parameter identifiability and evaluate potential risks to the conclusions drawn from the DDI study predictions. Sensitivity analysis was performed by

TABLE 1 | Input data for the PBPK model of T-DXd.

Parameter	Value	Source/Notes
T-DXd		
Phys chem and blood binding		
MW (g/mol)	150,000	Assumed as an average weight for IgG
Max DAR	8	[1]
Initial DAR distribution		
DAR8 (%)	90	
DAR6 (%)	10	
BP	0.55	Assumed
fu	1	
Main binding protein	HSA	
Distribution		
Model	Minimal PBPK	
FcRn binding K_D (mM)	2:1 stoichiometry	[16]
K_{D1}	1.39	Calibrated using measured value for trastuzumab
K_{D2}	5.56	Assumed $4 \times K_{D1}$ as shown for wild-type IgG [18]
PS_s (L/h)	0.001381	Predicted in Simcyp V21 using 2-pore hypothesis [19]
PS_l (L/h)	0.002795	
fu_{int}	1	Assumed
K_{up} (1/h)	0.0976	Optimized based on T-DXd PK profile (5.4 mg/kg) by parameter estimation, simultaneous estimation with FR
K_{rc1} (1/h)	3.13	Default values in Simcyp V21 [12]
K_{rc2} (1/h)	6.42	
σ_{vs}	0.908	Predicted in Simcyp V21 using 2-pore hypothesis [19]
σ_{VL}	0.163	
σ_L	0	Default values in V21
FR	0.598	Optimized based on T-DXd PK profile (5.4 mg/kg) by parameter estimation, simultaneous estimation with K_{up}
Elimination		
Model	Non-specific CL	
CL_{cat} (L/h)	0.0175	Assumed same as endogenous IgG [12]
Additional systemic clearance (L/h)	0.0017	
F_{rel}	0.5	Optimized based on DXd PK profile following release from T-DXd (5.4 mg/kg)
K_{dec}	0.004167	Default in Simcyp V21
DXd		
Phys chem and blood binding		
MW (g/mol)	493.48	Measured in vitro
Log P	2.3	
Compound type	Monoprotic acid	
pKa	10.64	
BP	0.6	
fu	0.02	

(Continues)

TABLE 1 | (Continued)

Parameter	Value	Source/Notes
Main binding protein	HSA	Assumed
Distribution		
Model	Full PBPK	
V_{SS} (L/kg) Method 3	0.20	Predicted; close to the estimated human V_{SS} (0.15 L/kg) from monkey after correcting for species difference in plasma protein binding
K_p scalar	1	
Elimination		
Model		
CYP3A4 CL_{int} (μ L/min/mg)	11.86	Total HLM CL_{int} = 12.5 (μ L/min/mg); CYP3A4 contribution in HLM metabolism = 94.9%, determined by chemical inhibition
Additional HLM CL_{int} (μ L/min/mg)	0.64	
fu_{mic}	0.87	Predicted
CL_R (L/h)	2.82	Allometrically scaled from monkey data
Hepatic transport		
CL_{PD} (mL/min/million cells)	0.010532	SIVA modeling using in vitro transporter data (Supporting Information)
OATP1B1 J_{max} (pmol/min/million cells)	109.3	
OATP1B1 K_M (μ M)	7.044	
OATP1B3 CL_{int} (μ L/min/million cells)	4.42	
P-gp CL_{int} (μ L/min/million cells)	13.81	
BCRP CL_{int} (μ L/min/million cells)	9.07	
P-gp REF	3.34	Calculated using abundances measured by the same technique in Caco-2 [20] and hepatocytes [21] (see Table S1 for details)
BCRP REF	2.5	

Abbreviations: BCRP, breast cancer resistance protein; BP, blood-to-plasma ratio; Caco, human colon adenocarcinoma cells (clone 2); CL, clearance; CL_{cat} , catabolic clearance; CL_{int} , intrinsic clearance; CL_{PD} , passive diffusion clearance; CL_R , renal clearance; CYP, cytochrome P450; DAR, drug antibody ratio; FcRn, neonatal Fc receptor; FR, recycling fraction of FcRn bound mAb; F_{rel} , fraction of catabolized payload released from the site of catabolism; fu , free fraction in plasma; fu_{int} , free fraction in interstitial space; fu_{mic} , free fraction in microsomes; HLM, human liver microsomes; HSA, human serum albumin; Ig, immunoglobulin; K_D , equilibrium dissociation constant; K_{D1} , equilibrium dissociation constant for 1:1 drug–FcRn complex; K_{D2} , equilibrium dissociation constant for 2:1 drug–FcRn complex; K_{dec} , degradation rate constant; Km, Michaelis Menten constant; K_p , tissue–plasma partition coefficient; K_{rec1} , recycling rate of FcRn–mAb 1:1 complex; K_{rec2} , recycling rate of FcRn–mAb 2:1 complex; K_{up} , endosomal uptake rate; Log P, log of the octanol–water partition coefficient for the neutral compound; mAb, monoclonal antibody; MW, molecular weight; OATP, organic anion transporting polypeptide; PBPK, physiologically based pharmacokinetic; P-gp, P-glycoprotein; PK, pharmacokinetic; pKa, acid dissociation constant; PS_p, permeability surface area product for large pores; PS_s, permeability surface area product for small pores; REF, relative expression factor; SIVA, Simcyp In Vitro Analysis toolkit; T-DXd, trastuzumab deruxtecan; V_{SS} , volume of distribution at steady state; σ_L , lymphatic reflection coefficient; σ_{vi} , vascular reflection coefficient for large pores; σ_{vs} , vascular reflection coefficient for small pores.

increasing or decreasing the contribution of CYP3A4 metabolism, uptake (OATP1B1 and OATP1B3), efflux (P-gp and BCRP), or renal clearance by up to 10-fold. The impact on the predicted DDI between unconjugated DXd and itraconazole or ritonavir following administration of T-DXd 5.4 mg/kg every 21 days was assessed.

3 | Results

3.1 | Model Development

The PBPK model accurately reproduced the observed PK parameters of T-DXd (5.4 mg/kg), with AUC_{tau} and C_{max} within 1.5-fold of the observed values (Table 2). Two different T-DXd F_{rel} values were tested. When 100% of DXd was released back to the blood ($F_{rel} = 100\%$), the DXd AUC_{tau} was overpredicted (Table 2). When only 50% of DXd was released back to the

blood ($F_{rel} = 50\%$), the recovery of the DXd PK profile was improved, and AUC_{tau} was predicted within 1.25-fold of the observed values (Table 2 and Figure 2). $F_{rel} = 50\%$ was used in subsequent simulations.

3.2 | Model Validation

The T-DXd and DXd profiles following administration of 6.4 mg/kg were well recovered by the model, with PK parameters predicted within 1.5-fold of the observed values (Table 2 and Figure 2). The simulated ritonavir and itraconazole DDI effects on DXd AUC were consistent with the observed data, with geometric mean ratios (GMRs) within 1.1-fold of the observed values (Table 3). The GMRs of C_{max} of DXd with ritonavir and itraconazole were slightly overpredicted, with GMRs within 1.5-fold of the observed values. The simulated results were comparable between $F_{rel} = 100\%$ and $F_{rel} = 50\%$

TABLE 2 | Simulated and observed geometric mean PK parameters of T-DXd and DXd following a 5.4 or 6.4 mg/kg dose every 21 days.

Dose (mg/kg)	$C_{\max}$		AUC_{τ}	
	(T-DXd: $\mu\text{g/mL}$; DXd: ng/mL)	(%)	(T-DXd: $\mu\text{g/mL}$; DXd: ng/mL)	(%)
5.4				
T-DXd				
Simulated	117	16.9	536	27.2
Observed 1 ^a	131	18.2	678	27.2
Observed 2 ^b	137	18.3	675	32.4
S/O1	0.893		0.790	
S/O2	0.854		0.794	
DXd with $F_{\text{rel}} = 100\%$				
Simulated	10.1	35.4	55.0	29.8
Observed 1 ^a	8.49	38.4	34.2	22.8
Observed 2 ^b	8.43	24.2	31.1	31.0
S/O1	1.18		1.61	
S/O2	1.19		1.77	
DXd with $F_{\text{rel}} = 50\%$				
Simulated	7.02	38.7	38.6	35.6
Observed 1 ^a	8.49	38.4	34.2	22.8
Observed 2 ^b	8.43	24.2	31.1	31.0
S/O1	0.827		1.13	
S/O2	0.833		1.24	
6.4				
T-DXd				
Simulated	138	16.9	632	27.2
Observed 1 ^c	163	28.0	744	30.5
Observed 2 ^d	152	22.7	668	29.5
S/O1	0.848		0.850	
S/O2	0.909		0.946	
DXd with $F_{\text{rel}} = 50\%$				
Simulated	8.40	38.8	45.7	35.6
Observed 1 ^c	8.80	49.3	40.4	36.0
Observed 2 ^d	11.7	32.1	38.5	28.0
S/O1	0.954		1.13	
S/O2	0.718		1.19	

Note: Observed data from Clinical Studies DS8201-A-A104 and DS8201-A-J101.

Abbreviations: AUC, area under the curve; $C_{\max}$, maximum plasma concentration; F_{rel} , fraction of catabolized payload released from the site of catabolism; PK, pharmacokinetic; S/O, simulated/observed; T-DXd, trastuzumab deruxtecan.

^aRitonavir cohort, Cycle 2.

^bItraconazole cohort, Cycle 2.

^cPart 2a.

^dPart 2c.

(Table S4). PK profiles showing the observed and predicted DXd concentrations in the presence of itraconazole and ritonavir are shown in Figure S1.

3.3 | DXd Pathway Mass Balance

The contribution of each elimination pathway in humans was evaluated following the simulation of a single dose of DXd directly (0.1 mg over a 0.5 h IV infusion) in the unconjugated DXd model (Table S5). Biliary clearance was the main route of elimination for DXd, accounting for 62.3% of the total elimination (Table S5). Renal elimination was also a notable pathway, contributing to 32.6% of the total elimination of DXd. Metabolism by CYP3A4 and other HLM pathways contributed minimally to the total elimination of DXd (4.8% by CYP3A4, 0.3% by others).

3.4 | Accounting for Risk of Uncertainty in DXd DDI Predictions

Sensitivity analyses of DXd clearance pathways with GMRs for DXd from DDI simulations following administration of T-DXd 5.4 mg/kg every 21 days in the presence and absence of ritonavir or itraconazole are shown in Figure 3, Table S6 (ritonavir), and Table S7 (itraconazole). A 10-fold increase or decrease in DXd CYP3A4 metabolism had a negligible impact on the predicted DDIs between T-DXd and ritonavir or itraconazole. A weak interaction ($1.25\text{-fold} \leq C_{\max}$ or AUC GMR $< 2\text{-fold}$) was predicted with ritonavir even when DXd uptake clearance was increased up to 10-fold, renal clearance was decreased up to 10-fold, or efflux clearance (into bile) was increased or decreased by up to 10-fold. No interaction was predicted with ritonavir when unconjugated DXd uptake clearance was reduced $\leq 10\text{-fold}$ or renal clearance was increased $\leq 10\text{-fold}$. Similar to the baseline model, no interaction was predicted with itraconazole even when DXd uptake clearance was reduced $\leq 10\text{-fold}$, efflux clearance was increased $\leq 10\text{-fold}$, or renal clearance was increased $\leq 10\text{-fold}$. A weak interaction ($1.25\text{-fold} \leq C_{\max}$ or AUC GMR $< 2\text{-fold}$) was predicted with itraconazole when DXd uptake clearance was increased $\leq 10\text{-fold}$, efflux clearance was decreased $\leq 10\text{-fold}$, or renal clearance was decreased by 10-fold. The baseline model for DXd generally gave the most accurate predictions of the observed DDI with T-DXd and ritonavir or itraconazole.

4 | Discussion

T-DXd is a HER2-targeted ADC composed of a humanized monoclonal antibody specifically targeting HER2, a cleavable tetrapeptide-based linker, and a potent topoisomerase I inhibitor payload (DXd). Non-clinical findings suggested that DXd is predominantly excreted as the unchanged form in the bile via OATP1B1/3-mediated hepatic uptake and excretion from the liver by P-gp and BCRP, with some metabolism by CYP3A and other CYPs. A clinical DDI study of T-DXd was conducted to evaluate the impact of inhibitors for CYP3A and OATP1B1/1B3 on the PK of DXd after T-DXd administration. This study revealed that the increase in DXd exposure was minimal (approximately 1.2-fold with ritonavir or itraconazole). In this study, a

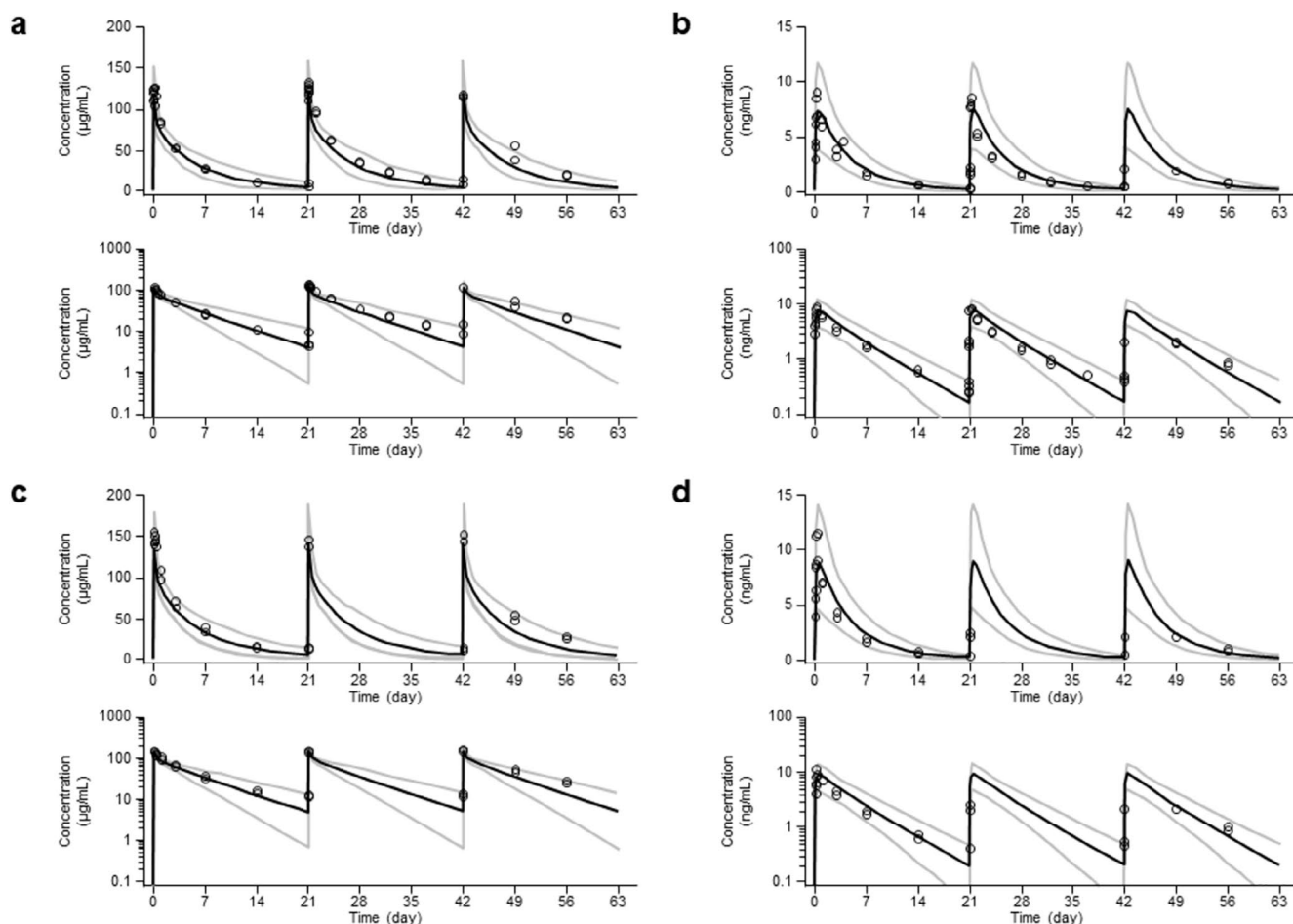


FIGURE 2 | Simulated mean (solid black line) and 5th and 95th percentile (gray lines) plasma concentrations of T-DXd (a, c) and released DXd (b, d) following a 5.4 mg/kg (a, b) or 6.4 mg/kg (c, d) dose every 21 days. The open circles represent the observed clinical data from the control arm of the drug interaction study (DS8201-A-A104) and the data from the dose-expansion cohort in DS8201-A-J101. Lower panels show the data with the y-axis on a logarithmic scale. T-DXd, trastuzumab deruxtecan.

PBPK model of T-DXd was developed and reported for the first time to account for the impact of these inhibitors on the PK of DXd and assist in the DDI risk assessment for subsequent DXd-containing ADCs.

The minimal ADC PBPK model was able to recover plasma concentration–time profiles and exposure levels of T-DXd and unconjugated DXd following administration of T-DXd 5.4 and 6.4 mg/kg with acceptable accuracy (Table 2). T-DXd $C_{\max}$ and AUC_{τ} predictions differed from the observed data by approximately 20%. The T-DXd F_{rel} was adjusted from 1 to 0.5 to reproduce the observed DXd AUC_{τ} . F_{rel} is a parameter that defines the proportion of DXd released into the bloodstream relative to the total amount of DXd bound to the administered ADC, and it affects the absolute exposure of DXd (AUC and $C_{\max}$). The reduction of F_{rel} indicates that within the PBPK model, not all DXd molecules liberated from the ADCs are released into the systemic circulation. This assumption seems reasonable, since it is likely that DXd released from T-DXd taken up by the liver is directly excreted into bile. In fact, the plasma exposure of DXd following administration of T-DXd to monkeys was approximately half that observed with direct administration of DXd (data on file).

The T-DXd PBPK model also reproduced the observed changes in the AUC of DXd in DDI studies with ritonavir and itraconazole. Both clinical data and PBPK simulations showed minimal increases in the DXd AUC_{τ} (Table 3). The PBPK model slightly overpredicted the DXd $C_{\max}$ GMRs compared with clinical data, in which no increases in $C_{\max}$ were observed. This discrepancy may be partly explained by clinical findings showing that the DXd $C_{\max}$ after multiple doses of T-DXd decreased by 24% compared with Cycle 1, likely due to reduced release from decreased numbers of HER2-expressing tumor cells after multiple doses of T-DXd [24]. The DDI study between T-DXd and ritonavir or itraconazole was designed as a single-sequence crossover study to compare Cycle 2 (ADC alone) versus Cycle 3 (ADC with inhibitor). When the inhibitor is present, the $C_{\max}$ in Cycle 3 may increase; however, this increase may not be apparent relative to the $C_{\max}$ value observed in Cycle 2 without the inhibitor, owing to the decreased number of HER2-expressing tumor cells. In the T-DXd PBPK model, the phenomenon leading to the decreased payload release was not considered, which may have led to the overprediction of the $C_{\max}$ GMR.

The impact of F_{rel} (1 vs. 0.5) on the prediction of DDI with ritonavir and itraconazole was also investigated (Table S4). The

TABLE 3 | Simulated and observed geometric mean (90% CI) AUC and $C_{\max}$ ratios of DXd following 5.4 mg/kg T-DXd in the presence and absence of ritonavir or itraconazole.

	AUC GMR ^a	(90% CI)	$C_{\max}$ GMR ^a	(90% CI)
Ritonavir DDI				
Simulated Japanese	1.24	(1.22–1.26)	1.24	(1.22–1.26)
Simulated cancer	1.32	(1.30–1.35)	1.33	(1.30–1.35)
Observed	1.22	(1.08–1.37)	0.987	(0.854–1.14)
Japanese S/O	1.01		1.26	
Cancer S/O	1.09		1.34	
Itraconazole DDI				
Simulated Japanese	1.20	(1.18–1.22)	1.19	(1.17–1.20)
Simulated cancer	1.20	(1.18–1.23)	1.19	(1.17–1.21)
Observed	1.18	(1.11–1.25)	1.04	(0.917–1.18)
Japanese S/O	1.02		1.14	
Cancer S/O	1.02		1.15	

Note: Observed data from Clinical Study DS8201-A-A104. Abbreviations: AUC, area under the curve; CI, confidence interval; $C_{\max}$, maximum plasma concentration; DDI, drug–drug interaction; GMR, geometric mean ratio; S/O, simulated/observed; T-DXd, trastuzumab deruxtecan.
^aCalculated using Cycle 3 (Days 1–17)/Cycle 2 (Days 1–17).

predicted GMRs of the DXd $C_{\max}$ and AUC values were the same regardless of the F_{rel} values. Because DDI primarily evaluates the effect on DXd released into the bloodstream, if F_{rel} remains the same in the presence and absence of an inhibitor, and if the DXd blood concentration is sufficiently low and within the linear range, it will not affect the DDI results.

A recognized limitation of the PBPK model for DXd is the absence of human PK data to verify the performance of the model when DXd is administered directly. Mass balance following different elimination pathways in humans was evaluated following administration of a single dose of DXd directly (0.1 mg administered over 0.5 h via IV infusion) in the unconjugated DXd model. The renal clearance, liver metabolism, and biliary clearance in humans were predicted to be 32.6%, 5.1% (4.8% by CYP3A4, 0.3% by others), and 62.3%, respectively (Table S5). The pre-clinical data indicated that at 24 h after IV administration of 1 mg/kg [¹⁴C]DXd to bile duct-cannulated rats, 19.8%, 71.5%, and 1.0% of the radioactivity were excreted in urine, bile, and feces, respectively. At 336 h after IV administration of 6.4 mg/kg [¹⁴C]T-DXd to monkeys, 86.0% of the radioactivity was recovered, with 18.7% in urine and 67.3% in feces [4]. No metabolite peaks were detected in the excreta of both species. The most abundant radioactive component was unchanged DXd. The predicted mass balance in humans based on the DXd model was consistent with the mass balance data from the pre-clinical species (rat and monkey).

To evaluate the impact of the estimated contribution of DXd clearance pathways on predicted DDIs with ritonavir and itraconazole, sensitivity analyses of each clearance parameter were conducted (Figure 3, Tables S6 and S7). In particular,

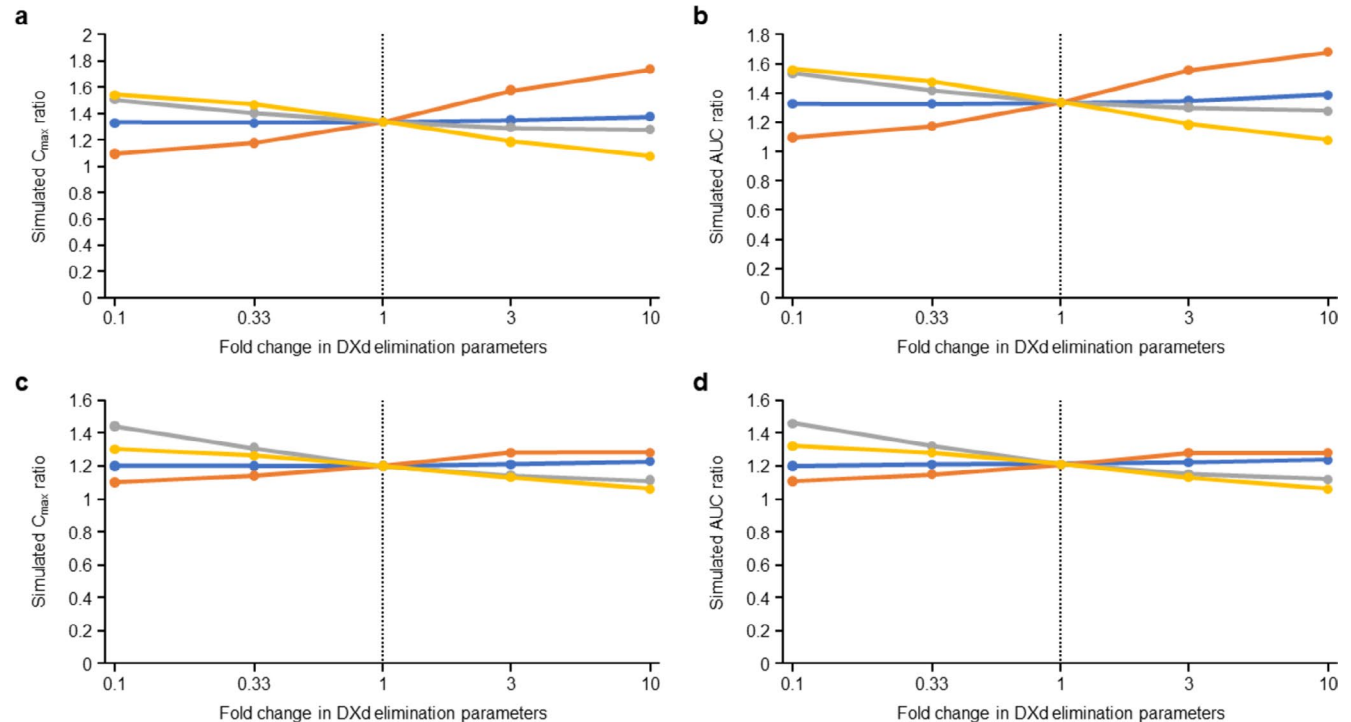


FIGURE 3 | Sensitivity analysis for the contribution of different elimination routes to interaction with itraconazole and ritonavir. Panels (a–d) represent the simulated DXd $C_{\max}$ (a, c) and AUC (b, d) geometric mean ratios following administration of T-DXd in the presence and absence of ritonavir (a, b) or itraconazole (c, d) with differing contributions of the elimination routes for DXd clearance. Blue, gray, orange, and yellow lines represent the impact of up to 10-fold increases or decreases in CYP3A4 metabolism, efflux clearance, uptake clearance, and renal clearance parameters, respectively. AUC, area under the curve; $C_{\max}$, maximum plasma concentration; CYP, cytochrome P450; T-DXd, trastuzumab deruxtecan.

changes in uptake clearance and renal clearance had some impact on the results of the DDI prediction. An increase or decrease in uptake clearance values resulted in an increase or decrease in the $C_{\max}$ and AUC ratios, while changes in renal clearance had the opposite effect. However, the magnitude of these changes was limited, and the classification as a weak interaction remained unchanged. In contrast, the mass balance was markedly altered when individual elimination clearances were varied (data not shown). Considering both the plausibility of the mass balance based on animal data and the reproducibility of the clinical DDI study results, the reliability of the current model can be considered reasonably high.

Another limitation of the PBPK model is that time-mediated drug disposition (TMDD) was not considered within the PBPK model, as the doses of interest were high enough that TMDD was saturated. The dose-escalation phase in Clinical Study DS8201-A-J101 established that area under the curve from 0 to last (AUC_{last}) increased proportionally with doses above 3.2 mg/kg [11]. Hence, extrapolation of the PBPK model to lower doses or ADCs with the same linker and payload but different antibody, where TMDD may be important (i.e., not saturated), is not appropriate. However, this PBPK model structure is considered sufficient for predicting DDI at doses where TMDD is saturated.

In conclusion, the PBPK model of T-DXd, constructed using nonclinical and clinical results obtained to date, appears to appropriately describe the PK characteristics of DXd and to reasonably reproduce the results of clinical DDI studies. This PBPK model is expected to be useful for DDI evaluation of subsequent DXd ADCs with the same payload and linker. The DXd model can be directly applied when evaluating new DXd-containing ADCs. To construct the ADC model, the minimum in vitro and preclinical data needed for a reliable prediction would include the K_D value and drug-to-antibody ratio distribution for FcRn. If the PK profiles of the ADC and DXd differ from those of T-DXd, the parameters of the ADC model (e.g., K_{up} , FR, and F_{rel}) must be adjusted to accurately represent these observed PK profiles. F_{rel} is thought to depend primarily on the extent of ADC uptake into target-specific and non-target tissues. Even for subsequent ADCs with the same payload and linker, F_{rel} may differ if tissue distribution differs; therefore, optimization of F_{rel} using clinical PK data from subsequent ADCs is necessary. In fact, the U.S. Food and Drug Administration review of datopotamab deruxtecan [25], a subsequent DXd-based ADC following T-DXd, noted that this PBPK analysis is applicable to evaluating DDI potential for DXd-based ADCs that share the same linker and payload as T-DXd. Moreover, this PBPK model will be widely useful for DDI prediction under different scenarios (such as different doses or inhibitors) and PK prediction in specific populations (such as in pediatric populations and patients with organ impairment).

Author Contributions

All authors wrote the manuscript and designed the research. A.W., K.L.G., K.-F.C., and I.G. performed the research. A.W., K.L.G., K.-F.C., M.D.S.M., and I.G. analyzed the data.

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Conflicts of Interest

A.W. and M.K. are employees of Daiichi-Sankyo Co. Ltd. K.-F.C., H.M.J., M.D.S.M., I.G., and K.L.G. are employees of Certara UK Limited and hold shares in Certara.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Methods S1.** Analysis of in vitro DXd transporter data for parameterization of the physiologically based pharmacokinetic model. **Table S1:** Calculated REF values for P-gp and BCRP between Caco-2 cells and hepatocytes. **Methods S2.** Method for setting ritonavir transporter K_i values for OATP1B1 and OATP1B3 and its validity. **Table S2:** Pharmacokinetics of pravastatin (0.1 mg) in the presence of ritonavir (100 mg). **Table S3:** Impact of the K_i values of ritonavir for OATP1B1 and OATP1B3 on the AUC and C_{max} ratios. **Figure S1:** Pharmacokinetic profiles showing the observed and predicted DXd concentrations in the presence of itraconazole (a) and ritonavir (b). **Table S4:** Comparison of simulated geometric mean (90% CI) AUC and C_{max} ratios of DXd following 5.4 mg/kg T-DXd in the presence and absence of ritonavir or itraconazole between F_{rel} of 100% and 50%. **Table S5:** Simulated pathway mass balance for unconjugated DXd following a single dose of 0.1 mg DXd. **Table S6:** Simulated geometric mean (90% CI) AUC and C_{max} ratios of DXd following a 5.4 mg/kg dose of T-DXd in the presence and absence of ritonavir with variation of the contribution of each DXd elimination process. **Table S7:** Simulated geometric mean (90% CI) AUC and C_{max} ratios of DXd following a 5.4 mg/kg dose of T-DXd in the presence and absence of itraconazole with variation of the contribution of each DXd elimination process.